Analyzing the Nonlocality of Sparse Autoencoder Features

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Abstract

Sparse autoencoders (SAEs) have emerged as a powerful tool for extracting interpretable features from the internal representations of large language models (LLMs). Motivated by an analogy with holographic duality in theoretical physics, where strongly correlated boundary degrees of freedom map to weakly coupled bulk fields of varying nonlocality, we introduce a gradient-based entropy measure that quantifies how nonlocal each SAE feature is in the input token space. For a given feature at a given layer, we compute the Jacobian of the feature activation with respect to all input token embeddings, normalize the resulting influence distribution, and evaluate its Shannon entropy. Applying this measure to Pythia-70m-deduped with pretrained SAEs across six transformer layers, we find that: (i) features exhibit a broad distribution of nonlocality; (ii) deeper layers produce systematically more nonlocal features; (iii) the entropy of individual features saturates with increasing context length, indicating a finite effective support; and (iv) SAE features generally have lower entropy than the reconstructed token vectors, supporting the view that sparse features provide a low-entropy decomposition of the residual stream.

1 Introduction

Large language models (LLMs) based on the transformer architecture achieve remarkable performance across a wide range of natural language tasks, yet their internal representations remain poorly understood. A central challenge in mechanistic interpretability is to decompose the dense, high-dimensional activations of a transformer into meaningful, human-interpretable units [Bricken et al., 2023, Templeton et al., 2024].

Sparse autoencoders (SAEs) offer a promising approach to this problem [Yun et al., 2021, Cunningham et al., 2023, Bricken et al., 2023, Templeton et al., 2024]. An SAE consists of a nonlinear encoder $f = E(y)$ that maps an internal activation vector $y \in \mathbb{R}^d$ to a higher-dimensional feature vector $f \in \mathbb{R}^M$ (with $M \gg d$), together with a linear decoder that approximately reconstructs the original activation. The encoder typically applies a ReLU nonlinearity,

$$f_a = \mathrm{ReLU}\Big(\sum_\mu W^{\mathrm{enc}}_{a\mu}\, y_\mu + b^{\mathrm{enc}}_a\Big), \tag{1}$$

where the sparsity of f is enforced through an L_1 penalty during training [Ranzato et al., 2007, Ng, 2011]. The decoder is a linear map,

$$\tilde{y}_\mu = \sum_a D_{\mu a}\, f_a\,, \tag{2}$$

satisfying $\tilde{y} \simeq y$. Although $\dim(f) > \dim(y)$, the sparsity constraint ensures that only a small number of features are active for any given input, so the SAE still performs a form of compression.

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In a causal transformer, the residual stream activation at position t and layer n is denoted $y_t^{(n)}$. The attention mechanism gives $y_t^{(n+1)} = \mathrm{Attn}_n(y_{t' \leq t}^{(n)})$, so that deeper layers can integrate information from progressively more input tokens. An SAE trained on a particular layer produces per-position feature vectors $f_t^{(n)} = E_n(y_t^{(n)})$, and the resulting features have been shown to correspond to interpretable concepts such as named entities, syntactic patterns, and semantic categories [Bricken et al., 2023, Templeton et al., 2024].

In this work, we introduce a gradient-based entropy measure that quantifies the *nonlocality* of each SAE feature in the input token space—that is, how many input tokens significantly influence a given feature activation. We perform a systematic numerical study across all layers and context lengths of a small transformer model (Pythia-70m-deduped), and we compare the nonlocality of individual features with that of the full token vector. Our main findings are:

- SAE features exhibit a broad distribution of nonlocality, with entropy ranging from less than one bit to over five bits.
- Deeper layers produce systematically more nonlocal features, consistent with encoding more abstract concepts.
- Feature entropy saturates with increasing context length, indicating a finite effective support for each feature.
- Individual features generally have lower entropy than the reconstructed token vector, supporting the interpretation that the residual stream is a superposition of features with different nonlocality scales.

2 Analogy with holographic duality

Our approach is motivated by a structural analogy between SAE features and bulk fields in the Anti-de Sitter/Conformal Field Theory (AdS/CFT) correspondence [Maldacena, 1999, Qi, 2018]. In holographic duality, a strongly correlated conformal field theory on the boundary of an anti-de Sitter spacetime is dual to a gravitational theory in the bulk. The bulk theory is weakly coupled in the semiclassical limit, and its local fields can be reconstructed as nonlocal functionals of boundary operators via the HKLL prescription [Hamilton et al., 2006]. Crucially, different bulk fields live at different radial depths: fields deeper in the bulk correspond to more nonlocal combinations of boundary operators. The Ryu–Takayanagi formula [Ryu and Takayanagi, 2006] further connects bulk geometry to boundary entanglement entropy, establishing a deep link between information-theoretic quantities and emergent spatial structure.

SAE features share three key structural properties with holographic bulk fields:

1. **Nonlocality.** Each SAE feature at an internal layer depends on multiple input tokens through the attention mechanism, just as a bulk field depends nonlocally on boundary operators.
2. **Approximate independence.** Well-trained SAE features are approximately uncorrelated for the natural distribution of inputs, analogous to the emergent locality of weakly coupled bulk fields.
3. **Sparsity.** Only a small number of features are active for any given input, paralleling the semiclassical regime where the number of bulk excitations is small.

This analogy suggests that different SAE features—corresponding to concepts at different levels of abstraction (e.g., individual words versus semantic categories such as "fruit" or "animal")—should exhibit different degrees of nonlocality in the input token space. If this is the case, one can assign to each feature a "depth" in an emergent radial dimension, with more abstract and nonlocal features sitting deeper. The primary goal of this paper is to test this picture quantitatively by measuring the nonlocality of SAE features across transformer layers.

3 Methodology

Motivated by the holographic analogy of Section 2, we seek a quantitative measure of how nonlocally each SAE feature depends on the input tokens. In holographic duality, the nonlocality of a bulk

field is characterized by how it is reconstructed from boundary operators; here, we characterize the nonlocality of a feature by how its activation depends on the input token embeddings, as measured through the gradient.

3.1 Gradient-based influence measure

Consider a feature f_a at token position t and layer n in a causal transformer. Due to the causal attention mask, $f_a(t, n)$ can depend on all input token embeddings $x_{\mu,t'}$ with $t' \leq t$, where $\mu = 1, \ldots, d_{\text{model}}$ indexes the components of the embedding vector. We define the *influence matrix* as the Jacobian of the feature activation with respect to the input embeddings:

$$I_{a,\mu}(t, n \mid t') = \frac{\partial f_a(t, n)}{\partial x_{\mu,t'}}, \qquad t' \leq t. \tag{3}$$

The squared Frobenius norm over the embedding index defines the *influence strength* of token t' on feature a:

$$J_a(t, n \mid t') = \sum_{\mu=1}^{d_{\text{model}}} [I_{a,\mu}(t, n \mid t')]^2. \tag{4}$$

By construction, $J_a(t, n \mid t') \geq 0$ for all t'. We choose the squared L_2 norm so that the total influence is additive:

$$J_{a,\text{tot}}(t, n) = \sum_{t' \leq t} J_a(t, n \mid t'). \tag{5}$$

3.2 Entropy of influence distribution

We normalize the influence strength to obtain a probability distribution over input token positions:

$$p(t') = \frac{J_a(t, n \mid t')}{J_{a,\text{tot}}(t, n)}. \tag{6}$$

The *influence entropy* is then the Shannon entropy of this distribution, measured in bits:

$$S_a(t, n) = -\sum_{t'} p(t') \log_2 p(t'). \tag{7}$$

The quantity 2^{S_a} gives the effective number of input tokens that significantly influence feature a. A feature that depends on only a single token has $S_a = 0$, while a feature that depends uniformly on all L tokens has $S_a = \log_2 L$.

In practice, we evaluate the entropy at the last token position $t = L$ of each input batch, since this position has access to all preceding tokens through the causal attention mask. For each feature, we average S_a over all batches in which the feature is active (i.e., $f_a >$ threshold). Alternative measures such as the inverse participation ratio $(\sum_{t'} p(t')^2)^{-1}$ are also possible but are not explored in this work.

3.3 Token vector entropy for comparison

To assess whether SAE features provide a "low-entropy decomposition" of the residual stream, we compare the influence entropy of individual features with that of the full reconstructed token vector. Recalling the decoder in Eq. (2),

$$\tilde{y}_\mu = \sum_a D_{\mu a} f_a = D_{\mu a_1} f_{a_1} + D_{\mu a_2} f_{a_2} + \cdots, \tag{8}$$

the Jacobian of the reconstructed vector with respect to the input embeddings is

$$\frac{\partial \tilde{y}_\nu(t, n)}{\partial x_{\mu,t'}} = \sum_a D_{\nu a} \frac{\partial f_a(t, n)}{\partial x_{\mu,t'}}. \tag{9}$$

We define the token vector influence as

$$R(t, n \mid t') = \sum_{\nu,\mu} \left(\frac{\partial \tilde{y}_\nu(t, n)}{\partial x_{\mu,t'}} \right)^2, \tag{10}$$

where the sum runs over both output component ν and input component μ. Normalizing $q(t') = R(t, n \mid t') / \sum_{t'} R(t, n \mid t')$ and computing the Shannon entropy yields the token vector entropy. Since $\tilde{y}$ is a superposition of features with potentially different nonlocality scales, we expect the token vector entropy to be higher than the entropy of typical individual features.

4 Experimental setup

Model. We use Pythia-70m-deduped [Biderman et al., 2023], a 6-layer causal transformer with hidden dimension $d_{\text{model}} = 512$, trained on the deduplicated Pile dataset.

Sparse autoencoders. We use pretrained SAEs from the `dictionary_learning` library [Marks et al., 2024], with $M = 32{,}768$ latent features per layer. Separate SAEs are trained for the residual stream output at each of the six layers (`resid_out_layer0` through `resid_out_layer5`).

Dataset. All analyses are performed on the WikiText-2 training set ($\sim$1M tokens). Input batches are constructed by taking contiguous sequences of tokens from the corpus.

Hyperparameters. The default batch size (context length) is $L = 64$ tokens. For the context-length dependence study (Section 5.5), we vary L from 8 to 128. A feature is considered active in a given batch if its activation exceeds a threshold of 0.2.

Leading features. We define *leading features* as those whose activation frequency (fraction of token positions where the feature is active) exceeds 0.1% across the dataset. At each layer, the top 500 features by activation frequency are selected for detailed analysis.

Gradient computation. For each batch, we perform a forward pass through the transformer and SAE with gradient tracking enabled on the input embeddings. The influence $\partial f_a(L, n)/\partial x_{\mu,t'}$ is computed for all active leading features a and all positions $t' \leq L$ using PyTorch `autograd`. The token vector influence (Eq. (10)) is computed using `torch.autograd.functional.jacobian` for the full Jacobian matrix.

5 Results

5.1 Feature sparsity statistics

Before analyzing nonlocality, we verify that the SAE features exhibit the expected sparsity structure. Figure 1 shows the distribution of the number of unique tokens that activate each feature at layer 0. The distribution is heavy-tailed: the vast majority of features activate on very few tokens, while a small number of features activate broadly across thousands of unique tokens.

To verify the approximate independence of features, we compute the correlation matrix $C_{ab} = \langle A^a A^b \rangle - \langle A^a \rangle \langle A^b \rangle$, where A^a is the binary activation indicator for feature a. As shown in Fig. 2, the matrix is approximately diagonal, confirming that the SAE features are approximately independent under the natural text distribution.

5.2 Influence distribution and entropy

For each active feature at the last token position, we compute the influence strength $J_a(t, n \mid t')$ across all input positions. Figure 3 shows the influence distribution for Feature 531 at layer 3, computed over 45 batches in which the feature was active. The influence is structured and non-uniform: certain input positions contribute significantly more than others, and the variance across batches is substantial but the mean profile shows clear structure.

The entropy S_a (Eq. (7)) is computed from the normalized influence distribution $p(t')$ for each batch, then averaged over batches in which the feature is active.

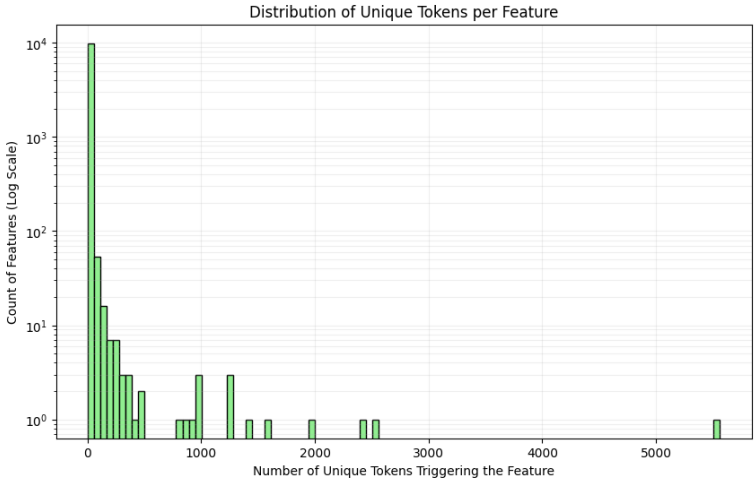

Figure 1: Distribution of the number of unique tokens activating each feature (layer 0). The vertical axis is on a logarithmic scale. The distribution is heavy-tailed, with most features activating on very few tokens and a small number activating on thousands of unique tokens.

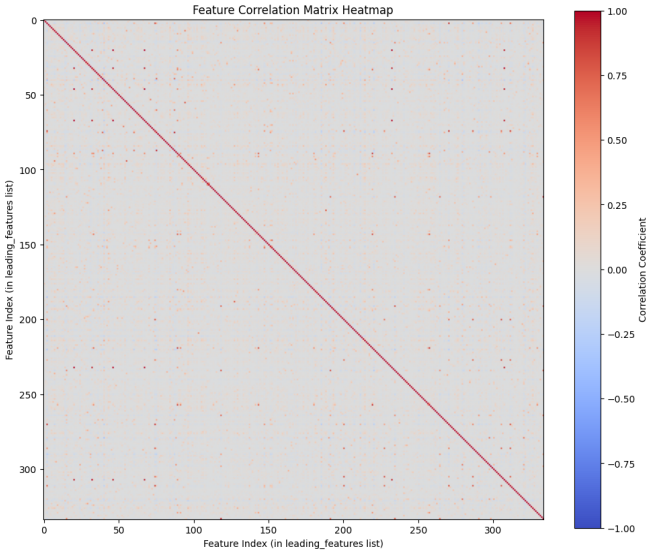

Figure 2: Feature correlation matrix C_{ab} for leading features at layer 3. The matrix is approximately diagonal, confirming approximate independence of SAE features under the natural text distribution. A small number of off-diagonal entries with elevated correlation are visible.

5.3 Entropy distribution across batches

For a given feature, the entropy S_a varies across batches depending on the specific input text. Figure 4 shows entropy histograms for several top features at layer 3. Each feature has a well-defined mean entropy with an approximately normal distribution. Importantly, different features have characteristically different mean entropies: Feature 6191 has a mean of 1.915 bits (highly local, with an effective support of $\sim 2^{1.9} \approx 4$ tokens), while Feature 20709 has a mean of 4.325 bits (highly nonlocal, with an effective support of $\sim 2^{4.3} \approx 20$ tokens). This confirms that the influence entropy captures a meaningful, feature-specific property.

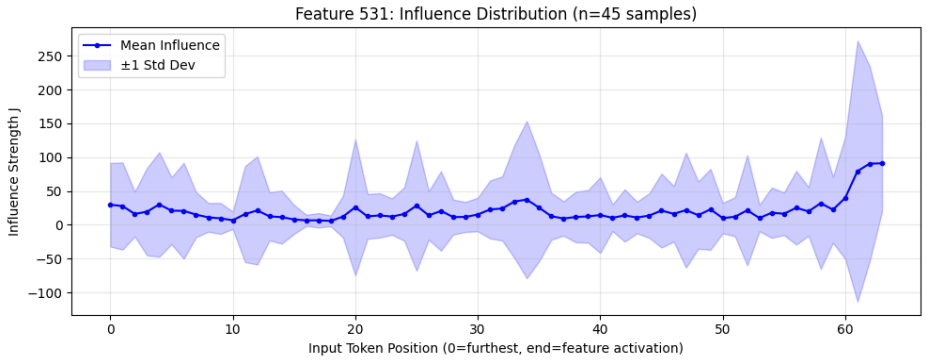

Figure 3: Influence distribution $J_a(t, n \mid t')$ for Feature 531 across 45 batches at layer 3. The horizontal axis is the input token position ($0 =$ furthest from the feature activation, end $=$ feature position). The blue curve shows the mean influence with ± 1 standard deviation (shaded). The feature shows structured, non-uniform dependence on input tokens, with elevated influence near the feature position.

5.4 Depth dependence of entropy

A key prediction of the holographic analogy is that features at deeper layers should be more nonlocal, since deeper layers have access to more tokens through the causal attention mechanism and encode more abstract concepts.

Figure 5 shows the influence entropy as a function of layer depth. Each gray dot represents one feature; colored lines track selected features across layers. The red dashed line shows the token vector entropy, and the blue dashed line shows the average entropy of the top-20 features at each layer. The average feature entropy increases monotonically with depth.

Figure 6 provides another view: histograms of the average feature entropy at each layer. The distribution shifts progressively to higher entropy at deeper layers. Quantitatively, the mean entropies (mean $\pm$ standard deviation) are: layer 0: 0.898 ± 0.618 bits (71 features); layer 1: 1.859 ± 0.880 bits (43 features); layer 2: 2.484 ± 0.922 bits (116 features); layer 3: 3.163 ± 0.842 bits (225 features); layer 4: 3.705 ± 0.809 bits (211 features). The number of leading features also increases with depth, reflecting the increasing complexity of representations. The trend continues at layer 5, where the mean entropy for 500 leading features is 4.054 bits (see Section 5.6).

5.5 Context length dependence

Figure 7 shows the influence entropy as a function of context length (batch size) for features at layer 2. As the context length increases, feature entropies initially grow but saturate beyond approximately 40–60 tokens. The saturation values are well below the maximum possible entropy $\log_2 n$ (shown as a dashed line), confirming that each feature has a finite effective support in token space. The saturation value is feature-dependent and reflects the characteristic nonlocality scale of each feature: some features saturate at ~ 1 bit regardless of context length, while others reach ~ 4 bits.

5.6 Feature versus token vector entropy

Finally, we compare the influence entropy of individual SAE features with that of the reconstructed token vector $\tilde{y}$. Figure 8 shows the averaged entropy versus activation probability for 500 leading features at layer 5, with a mean feature entropy of 4.054 bits. In per-batch comparisons (e.g., Fig. 5), the token vector entropy is indicated by a dashed red line. At layer 4, the token vector entropy is approximately 4.435 bits, with most individual features falling below this value. At layer 1, the token vector entropy is approximately 2.566 bits, and again most features have lower entropy.

These results are broadly consistent with the interpretation that the token vector is a superposition of features with different nonlocality scales, and that decomposing into SAE features yields a "low-entropy decomposition." However, the effect is not perfectly consistent across all features:

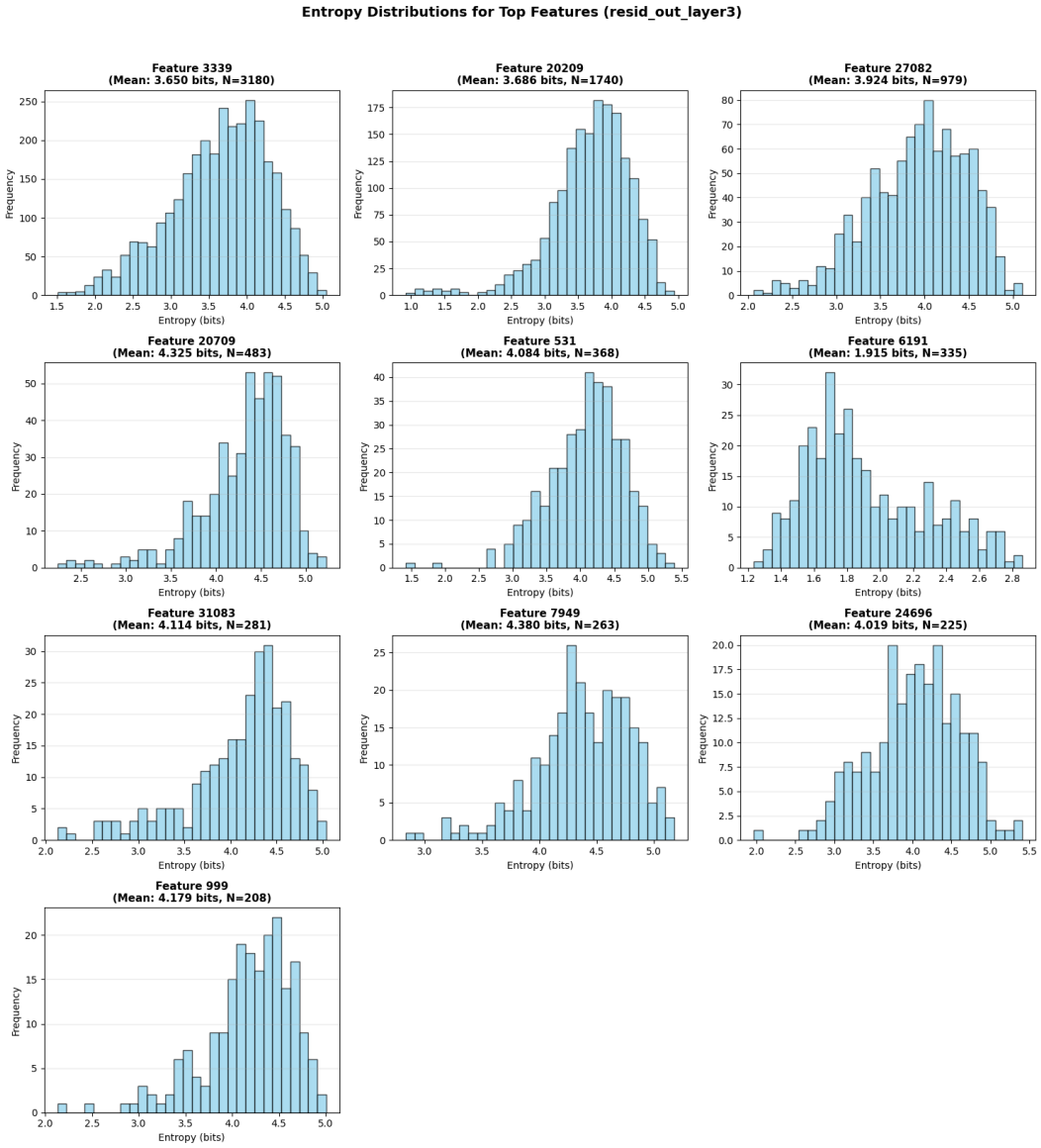

Figure 4: Entropy distributions across batches for top features at layer 3. Each panel shows the histogram of S_a over all batches where the feature is active, with the mean entropy and number of samples N indicated. Mean entropies range from 1.915 bits (Feature 6191, highly local) to 4.325 bits (Feature 20709, highly nonlocal).

some individual features can have entropy comparable to or exceeding that of the token vector, particularly at deeper layers where both feature and token vector entropies are high. A more systematic understanding of which features violate this pattern remains an open question.

6 Discussion and conclusion

We have introduced a gradient-based entropy measure that quantifies the nonlocality of sparse autoencoder features in the input token space of a transformer language model. Our main findings are:

1. **Nonlocality hierarchy.** SAE features exhibit a broad distribution of influence entropies, ranging from highly local ($\sim$1 bit, effectively depending on one or two tokens) to highly

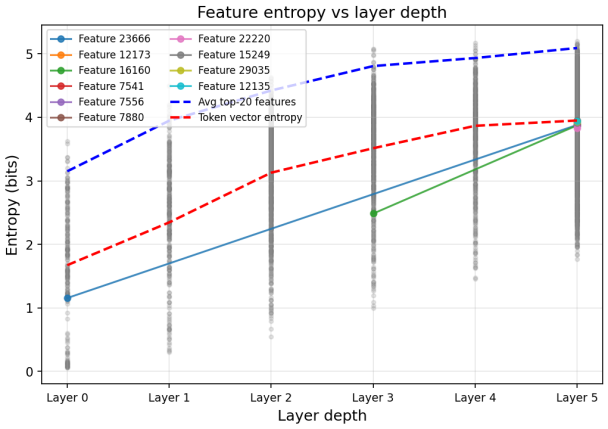

Figure 5: Feature entropy versus layer depth. Each gray dot is one feature; colored lines track selected features across layers. The red dashed line shows token vector entropy; the blue dashed line shows the average entropy of the top-20 features. Average entropy increases monotonically with layer depth, consistent with deeper layers encoding more nonlocal concepts.

nonlocal ($\sim$5 bits, effectively depending on $\sim$30 tokens). This supports the picture that SAE features capture concepts at varying levels of abstraction.

2. **Depth dependence.** Average feature entropy increases monotonically with layer depth, from 0.898 bits at layer 0 to 3.705 bits at layer 4. This is consistent with deeper layers encoding more abstract and context-dependent concepts, and is reminiscent of the radial dimension in holographic duality, where more nonlocal boundary operators correspond to bulk fields deeper in the geometry.
3. **Context length saturation.** Feature entropy saturates with increasing context length, indicating that each feature has a characteristic nonlocality scale that is intrinsic to the feature rather than determined by the available context window.
4. **Low-entropy decomposition.** Individual features generally have lower influence entropy than the full reconstructed token vector, supporting the view that SAE features decompose the residual stream into components with more structured dependence on the input.

Several open questions merit further investigation. First, alternative measures of nonlocality such as the participation ratio or Rényi entropies could provide complementary information about the structure of the influence distribution. Second, it would be valuable to establish criteria for when a feature extraction is "good" in terms of the entropy distribution it produces—for example, whether minimizing the average feature entropy subject to reconstruction accuracy leads to more interpretable features. Third, it remains to be seen whether the patterns we observe are universal across model scales and architectures. Fourth, cross-layer SAEs that share features across multiple layers [Shi et al., 2025, Bussmann et al., 2025] could reveal dynamical nonlocality in both the token and layer dimensions, potentially giving a richer picture of the emergent geometry. Fifth, the logical relations between features at different abstraction levels remain poorly understood, and recent work has highlighted potential pitfalls such as feature splitting, absorption, and hedging in narrow SAEs [Chanin et al., 2024, 2025].

More broadly, we believe that analyzing LLMs from the angle of information theory and emergent geometry is a fruitful direction. The holographic analogy, while imperfect, provides a useful conceptual framework for thinking about the structure of learned representations. The entropy measure introduced here is a first step toward making this analogy quantitative, and we hope it will inspire further connections between theoretical physics and the interpretability of neural networks.

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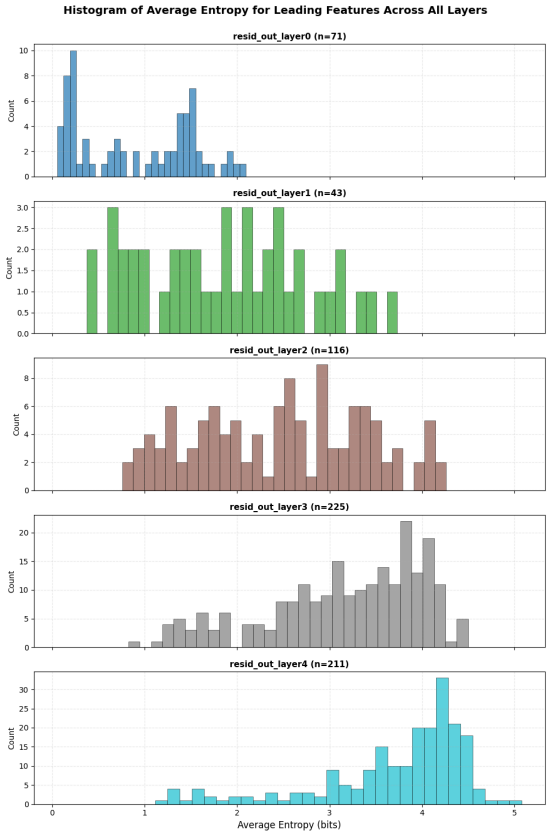

Figure 6: Histograms of average feature entropy for leading features across layers 0–4. The number of leading features at each layer is indicated in parentheses. The distribution shifts to higher entropy at deeper layers, consistent with more abstract and nonlocal representations.

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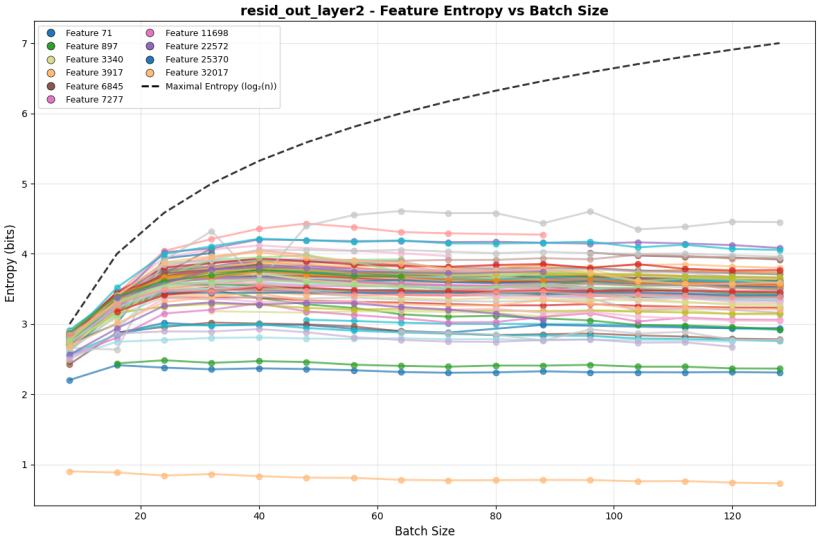

Figure 7: Feature entropy versus context length (batch size) for layer 2. Each curve corresponds to one leading feature. The dashed line shows the maximum possible entropy $\log_2 n$. Feature entropies saturate well below the maximum, indicating finite effective support. Highlighted features in the legend illustrate the range of saturation values.

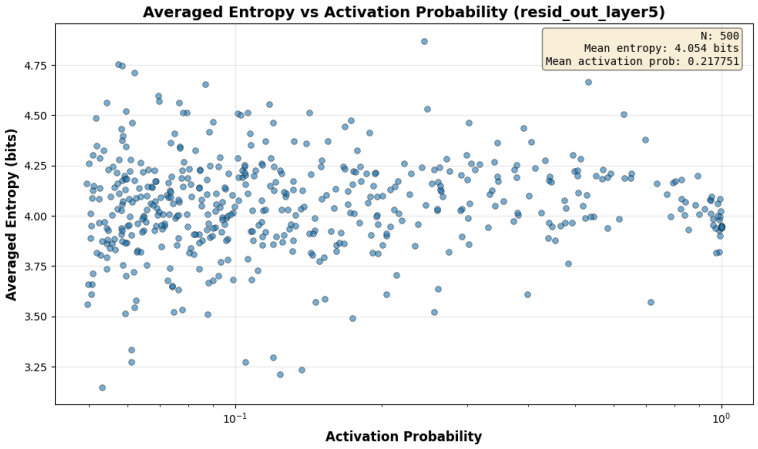

Figure 8: Averaged entropy versus activation probability for 500 leading features at layer 5. The mean entropy is 4.054 bits. Entropy is broadly distributed and does not depend strongly on activation frequency.

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A Details of the gradient computation

The influence matrix in Eq. (3) is computed using PyTorch automatic differentiation. For each batch of L tokens, we perform a forward pass through the transformer with gradient tracking enabled on the input embeddings. The embedding layer is temporarily replaced with a fixed tensor to ensure that the input embeddings are leaf variables in the computation graph, so that $\partial f_a / \partial x_{\mu,t'}$ is well defined.

For feature influences, we compute $\partial f_a(L,n)/\partial x_{\mu,t'}$ via `torch.autograd.grad` with `retain_graph=True` for each active feature a at the last token position. The squared norm (Eq. (4)) is computed from the resulting gradient tensor of shape $[1, L, d_{\text{model}}]$. For token vector influences, we use `torch.autograd.functional.jacobian` to compute the full Jacobian $\partial \tilde{y}_\nu(L,n)/\partial x_{\mu,t'}$ of shape $[d_{\text{model}}, L, d_{\text{model}}]$ in a single vectorized call. The token influence $R(t, n \mid t')$ is then obtained by summing the squared Jacobian over the output and input component indices.

All experiments were conducted on Apple M-series hardware using the MPS backend, with typical processing times of $\sim$0.5 seconds per batch for feature influences and $\sim$2 seconds per batch for the full Jacobian computation.